Coding Assignment 5 | DATA 5600

Categorical Variables

Kolbee Morris

In [146…

```
# load packages here
import pandas as pd
import seaborn as sns
from matplotlib import pyplot as plt
import statsmodels.api as sm
from statsmodels.stats.anova import anova_lm
```

Data and Description

Note that for the sake of length for this homework assignment, I am not having you check the model assumptions. You certainly can, if you would like, and in "real life" you would definitely need to do this prior to any statistical inference.

Macroeconomists often speculate that life expectancy is linked with the economic well-being of a country. Macroeconomists also hypothesize that Organisation for Economic Co-operation and Development (OECD) (an international think tank charged with promoting policies that will improve global social and economic well-being) members will have longer life expectancy. To test these hypotheses, the LifeExpectancy.txt data set (found on Canvas) contains the following information:

Variable	Description
LifeExp	Average life expectancy in years
Country	Country name
Group	Is the country a member of OECD, Africa, or other?
PPGDP	Per person GDP (on the log scale)

The Group variable indicates if the country is a member of the OECD, a member of the African continent, or belonging to neither group (other). **Note that the Country variable is just for your reference - you will not use this variable in your model.**

Download LifeExpectancy.txt, and put it in the same folder this Jupyter Notebook.

0. Replace the text "< YOUR NAME HERE >" (above) with your full name.

1. Read in the data set, call it "life" and remove the "Row" column. Print the first few rows of the data.

In [147…
```
life = pd.read_table("LifeExpectancy.txt", delimiter = ' ')
life = life.drop('Row', axis=1)
```

In [148…
```
life.head()
```

Out[148…

	Country	Group	PPGDP	LifeExp
0	Albania	other	8.209907	75.31
1	Anguilla	other	9.528801	79.19
2	Argentina	other	9.122831	77.72
3	Armenia	other	8.016549	74.54
4	Aruba	other	10.036772	79.80

2. Explore the continuous data by (1) using .describe(), and (2) histograms. Comment on the output.

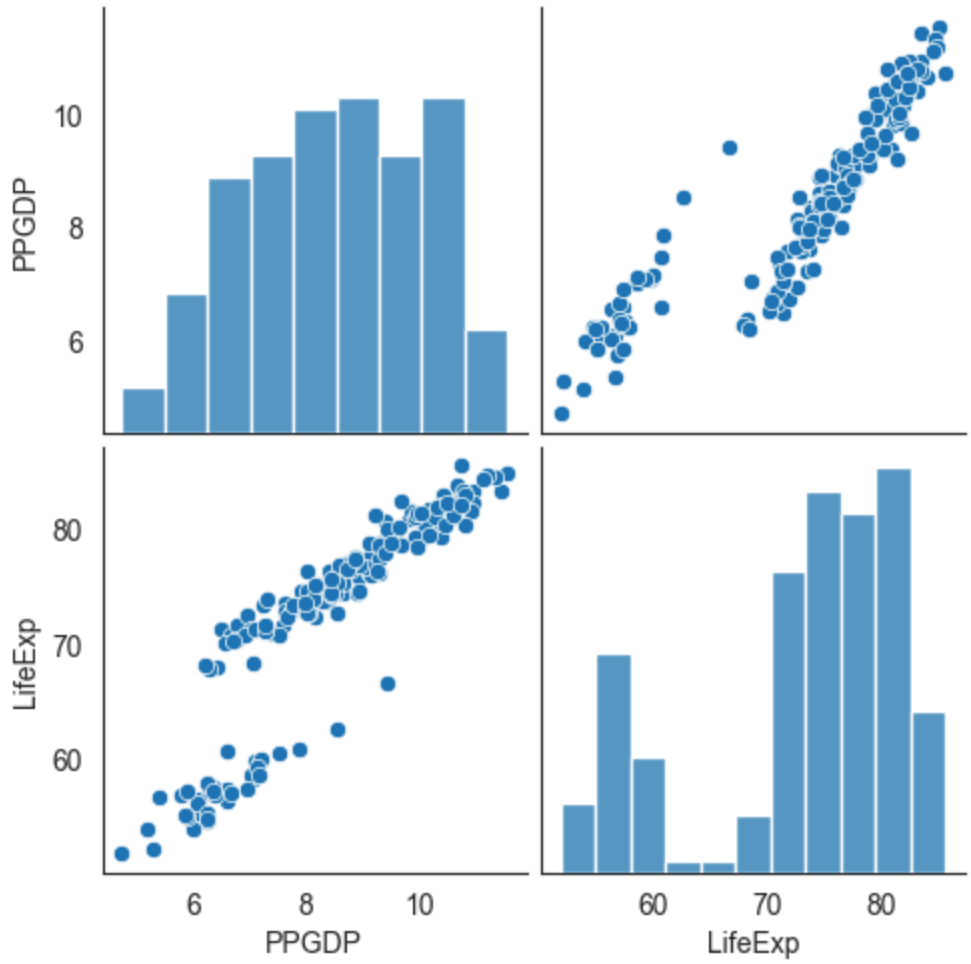
In [149…
```
life.describe()
```

Out[149…

	PPGDP	LifeExp
count	181.000000	181.000000
mean	8.491746	73.136630
std	1.599368	8.948747
min	4.743191	51.860000
25%	7.156645	70.810000
50%	8.541535	75.290000
75%	9.843674	79.800000
max	11.562624	85.620000

In [150…
```
sns.pairplot(life)
```

Out[150…
```
<seaborn.axisgrid.PairGrid at 0x11c7f1010>
```

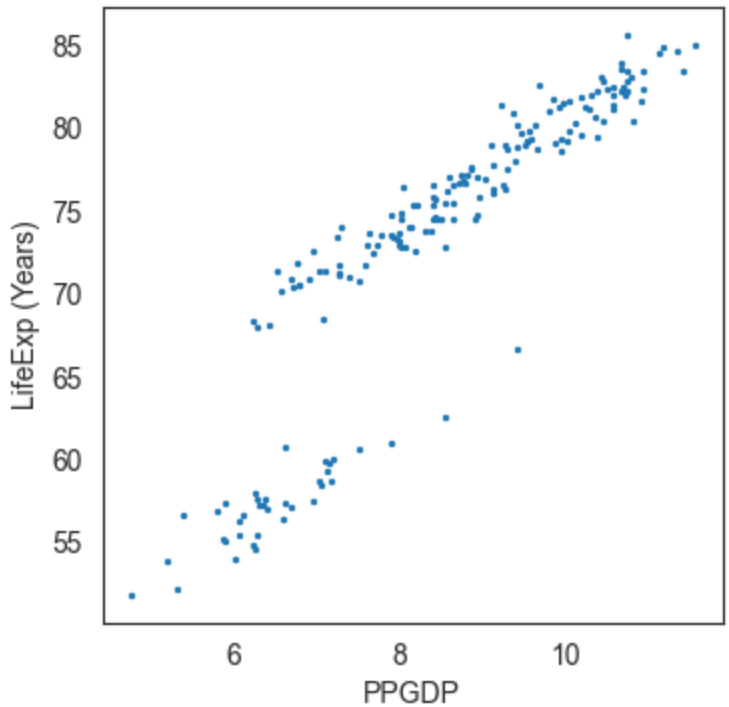


It looks like the Histograms show a little left skew in both of them. The plot charts look like you can fit two strong, positive, linear lines in the data set.

3. Create and print a scatterplot with the response on the y-axis and the other continuous variable on the x-axis. Comment on the the relationship between these two variables.

In [151…

```
fig = plt.figure(figsize = (4, 4))
plt.scatter(x = life['PPGDP'],
            y = life['LifeExp'],
            s = 2)
plt.xlabel("PPGDP")
plt.ylabel("LifeExp (Years)")
plt.show()
```



It looks like the relationship between the two varibles are strong, positive, and linear. You can fit two lines on the chart but over all looks like there is a good correlation.

4. Explore the categorical data using .value_counts(). Answer the following: (a) How many levels are there? (b) How many parameters with this variable add to the regression model? (c) Are any levels sparse (very small group sizes)?

In [152…
```
life['Group'].value_counts()
```

Out[152…
```
Group
other     113
africa     37
oecd       31
Name: count, dtype: int64
```

(a) There are 3 Levels. Other, Africa, and oecd.

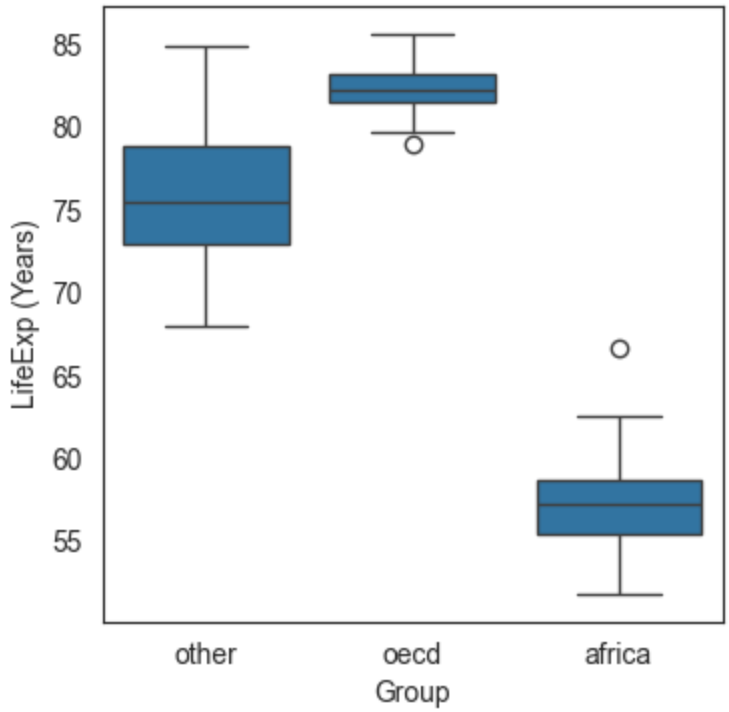
(b) This variable will add two parameters to the regression model.

(c) There are not any groups that are very sparse.

5. Create and print a boxplot with the response on the y-axis and the categorical variable on the x-axis. Comment on the the relationship between these two variables (Hint: use seaborn).

In [153…
```
plt.figure(figsize = (4, 4))
sns.boxplot(x = life['Group'],
            y = life['LifeExp'],
            data = life)
plt.xlabel("Group")
```

```
plt.ylabel("LifeExp (Years)")
plt.show()
```

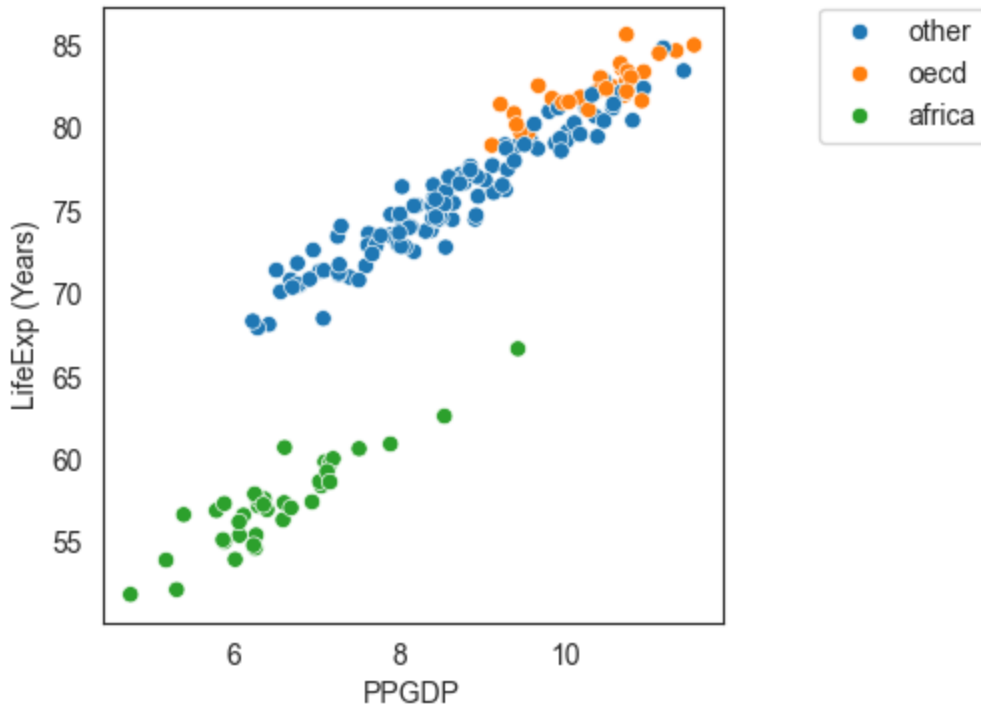


The other group is pretty widespread. but shows that the majority of the points are around 75 years life expectancy. The oecd group has the least variety in the boxplot and averages around a lifespan of 82 years old. with one plot being outside the wisker on the lower side. The Africa Group has a low life expectancy between their countries averaging around a 57 age expectancy with one country outside the boxplot on the higher side.

6. Create and print a color-coded scatterplot using all of the variables that will be in your model. Hint: plot the response on the y-axis, the other continuous variable on the x-axis, and color the points by the categorical variable.

In [154…

```
plt.figure(figsize = (4, 4))
sns.scatterplot(data = life,
                x = 'PPGDP',
                y = 'LifeExp',
                hue = 'Group')
plt.xlabel('PPGDP')
plt.ylabel('LifeExp (Years)')
plt.legend(loc = 'upper right', bbox_to_anchor = (1.45, 1.02))
plt.show()
```



7. Write out the general/theoretical model (using Greek letters/parameters) that you are thinking about applying to this data set (you will not write out the fitted model using coefficients, because you have not fit a model yet;)). Remember, you will need to use dummy variables for Group. **USE "other" AS THE BASELINE CATEGORY**. Use variable names that are descriptive (not y, x_1, etc.).

$$\text{LifeExp}_i = \beta_0 + \beta_1(\text{OECD}_i) + \beta_2(\text{Africa}_i) + \epsilon_i$$

8. Do you have enough rows to fit the model you wrote above? Justify by showing your calculations.

In [155…

```
dummies = pd.get_dummies(life['Group'], prefix='Group', dtype=int)

# 2. Select the specific columns for your model: 'Group_oecd' and 'Group_afr
# We join these back to the dataframe or use them directly
X = dummies[['Group_oecd', 'Group_africa']]

# 3. Add the intercept (beta_0)
X = sm.add_constant(X)

# 4. Define y
y = life['LifeExp']

# 5. Fit the OLS model
model = sm.OLS(y, X).fit()

# 6. Print the summary
print(model.summary())
```

```
                            OLS Regression Results
==============================================================================
==
Dep. Variable:                LifeExp   R-squared:                       0.8
67
Model:                            OLS   Adj. R-squared:                  0.8
65
Method:                 Least Squares   F-statistic:                     57
7.9
Date:                Mon, 23 Feb 2026   Prob (F-statistic):           1.42e-
78
Time:                        19:03:26   Log-Likelihood:                -470.
72
No. Observations:                 181   AIC:                             94
7.4
Df Residuals:                     178   BIC:                             95
7.0
Df Model:                           2
Covariance Type:            nonrobust
==============================================================================
====
                   coef    std err          t      P>|t|      [0.025      0.
975]
------------------------------------------------------------------------------
----
const           75.7634      0.309    244.997      0.000      75.153      7
6.374
Group_oecd       6.5292      0.666      9.796      0.000       5.214
7.844
Group_africa   -18.3201      0.623    -29.423      0.000     -19.549     -1
7.091
==============================================================================
==
Omnibus:                        1.511   Durbin-Watson:                   2.0
04
Prob(Omnibus):                  0.470   Jarque-Bera (JB):                1.3
28
Skew:                           0.209   Prob(JB):                        0.5
15
Kurtosis:                       3.031   Cond. No.                         3.
17
==============================================================================
==

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is corre
ctly specified.
```

yes I have enough rows to fit the model. 181 observations minus the 3 parameters are more than enough to fit the model.

9. Code indicator/dummy variables for Group.

In [156…

```
life_dummy = pd.get_dummies(life,
                            columns = ['Group'],
```

```
                                dtype = int)
life_dummy.head()
```

Out[156…

	Country	PPGDP	LifeExp	Group_africa	Group_oecd	Group_other
0	Albania	8.209907	75.31	0	0	1
1	Anguilla	9.528801	79.19	0	0	1
2	Argentina	9.122831	77.72	0	0	1
3	Armenia	8.016549	74.54	0	0	1
4	Aruba	10.036772	79.80	0	0	1

10. Fit a multiple linear regression model to the data (no transformations) using "other" as the baseline for group. Print a summary of the results.

In [157…

```
y = life_dummy['LifeExp']
X = sm.add_constant(life_dummy[['Group_africa', 'Group_oecd']])
mod = sm.OLS(y, X)
res = mod.fit()
res.summary()
```

Out[157…

OLS Regression Results

Dep. Variable:	LifeExp	**R-squared:**	0.867
Model:	OLS	**Adj. R-squared:**	0.865
Method:	Least Squares	**F-statistic:**	577.9
Date:	Mon, 23 Feb 2026	**Prob (F-statistic):**	1.42e-78
Time:	19:03:26	**Log-Likelihood:**	-470.72
No. Observations:	181	**AIC:**	947.4
Df Residuals:	178	**BIC:**	957.0
Df Model:	2		
Covariance Type:	nonrobust		

	coef	std err	t	P>|t|	[0.025	0.975]
const	75.7634	0.309	244.997	0.000	75.153	76.374
Group_africa	-18.3201	0.623	-29.423	0.000	-19.549	-17.091
Group_oecd	6.5292	0.666	9.796	0.000	5.214	7.844

Omnibus:	1.511	**Durbin-Watson:**	2.004
Prob(Omnibus):	0.470	**Jarque-Bera (JB):**	1.328
Skew:	0.209	**Prob(JB):**	0.515
Kurtosis:	3.031	**Cond. No.**	3.17

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

11. Briefly interpret the intercept (like we did in class). **Note that you will need to use the word "average" (or similar) twice since you are predicting an average already.** You will need to do this (say average twice) here and with the questions that follow when interpreting.

The Intercept is 75.76 it represents the average life expectancy for the Other group, and this serves as the average predicted value of that group.

12. Briefly interpret the coefficient for PPGDP (log scale) (like we did in class). You do not need to un-transform anything - you can just write something like "per person GDP (log scale)" in your response.

For every one unit increase in PPGDP (log scale), the average life expectancy is expected to increase by 2.87 years on average if all other varibles are held constant.

13. For equal per person GDP (log scale), how does life expectancy change for countries that are members of the OECD compared to countries that are on the African continent? Show how you obtained this number, and briefly interpret this number (like we did in class).

In [158…

```
y = life_dummy['LifeExp']
X = sm.add_constant(life_dummy[['PPGDP', 'Group_africa', 'Group_oecd']])
mod = sm.OLS(y, X)
res = mod.fit()
res.summary()
```

Out[158…

OLS Regression Results

Dep. Variable:	LifeExp	**R-squared:**	0.986
Model:	OLS	**Adj. R-squared:**	0.986
Method:	Least Squares	**F-statistic:**	4080.
Date:	Mon, 23 Feb 2026	**Prob (F-statistic):**	4.48e-163
Time:	19:03:26	**Log-Likelihood:**	-268.31
No. Observations:	181	**AIC:**	544.6
Df Residuals:	177	**BIC:**	557.4
Df Model:	3		
Covariance Type:	nonrobust		

	coef	std err	t	P>|t|	[0.025	0.975]
const	50.9579	0.653	78.070	0.000	49.670	52.246
PPGDP	2.8769	0.075	38.470	0.000	2.729	3.024
Group_africa	-12.2943	0.257	-47.789	0.000	-12.802	-11.787
Group_oecd	1.5298	0.254	6.019	0.000	1.028	2.031

Omnibus:	0.441	**Durbin-Watson:**	2.080
Prob(Omnibus):	0.802	**Jarque-Bera (JB):**	0.333
Skew:	0.105	**Prob(JB):**	0.847
Kurtosis:	3.015	**Cond. No.**	75.0

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

For countries with an equal PPGDP (log Scale) the average life expectancy for a country from a oced is predicted to be 13.82 years higher on average than the average life in a country in the africa group

14. Briefly interpret the 95% confidence interval for I(Group=Africa).

We are 95% confident that countries with equal PPGDP (log scale) the average life expectancy is between 12.802 and 11.787 years lower on average than the average life expectancy for countries in the other group.

15. Use the `anova_lm` function from statsmodels to conduct a hypothesis test that tests if Group as a whole has a significant effect on LifeExp. What do you conclude from the result of the test? Hint: you will need to create another linear model and compare it with the one you made previously.

In [159…

```
res_reduced = sm.OLS(y, X[['const', 'PPGDP']]).fit()
anova_lm(res_reduced, res)
```

Out[159…

	df_resid	ssr	df_diff	ss_diff	F	Pr(>F)
0	179.0	2858.842765	0.0	NaN	NaN	NaN
1	177.0	205.477622	2.0	2653.365143	1142.814545	6.409470e-102

Since the P value is $6.409470 \times 10^{-102}$ it is lower than 0.05 so we reject the null hypothesis. this means that PPGDP has a statistically significant effect on life expectancy.

16. Create a 95% prediction interval for the life expectancy of a country in the OECD with an average per person GDP (log scale) of 9.5. Print the result, and briefly interpret this interval (like we did in class). (Use the `get_prediction` function on the `OLS` object.)

In [160…

```
pred = res.get_prediction([1, 9.5, 0, 1])
result = pred.summary_frame(alpha = 0.05).iloc[:, [0, 2, 3]]
print(result)
```

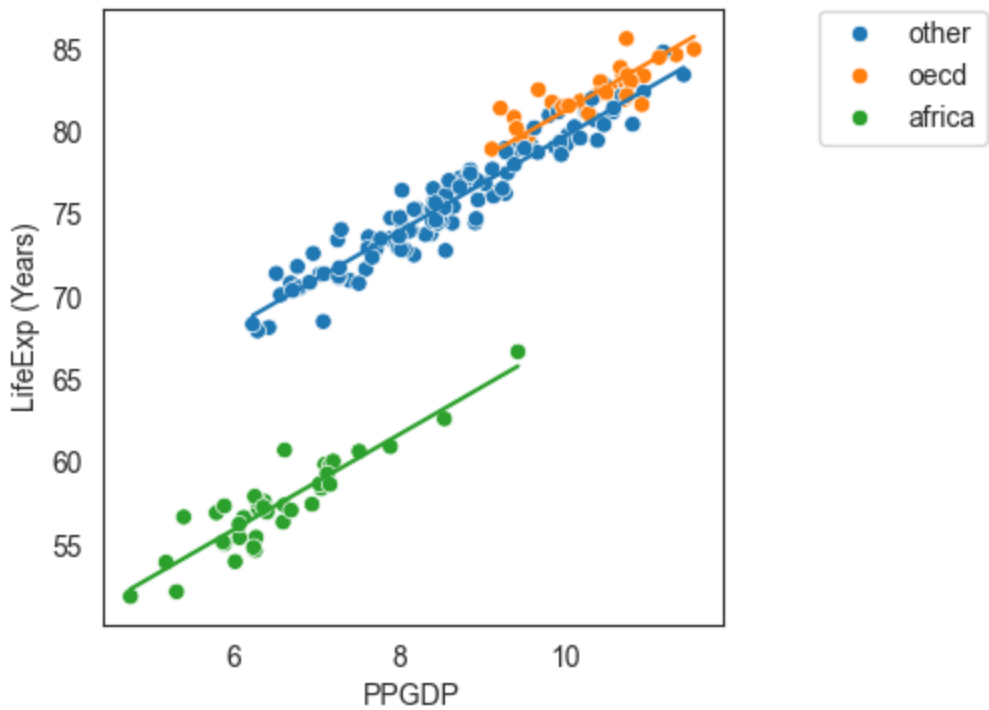
```
       mean  mean_ci_lower  mean_ci_upper
0  79.818286      79.415851       80.22072
```

We are 95% confident that the life expectancy for a specific country in OCED with a PPGDP (Log value) of 9.5 will fall between 79.415 and 80.220 years.

17. Plot the fitted model on the scatterplot with the two continuous variables on the axes, colored by the categorical variable. Hint: you should have 3 different lines on your plot, and you will *not* need to have different line types or point shapes (you *will* need to have different colors).

In [161…

```
sns.set_style("white")
plt.figure(figsize = (4, 4))
sns.scatterplot(data = life,
                x = 'PPGDP',
                y = 'LifeExp',
                hue = 'Group')
sns.lineplot(x = life['PPGDP'],
             y = res.fittedvalues,
             hue = life['Group'],
             legend = False)
plt.xlabel('PPGDP')
plt.ylabel('LifeExp (Years)')
plt.legend(loc = 'upper right', bbox_to_anchor = (1.45, 1.02))
plt.show()
```



18. Briefly summarize what you learned from this analysis *to a non-statistician*. Write a few sentences about (1) the purpose of this data set and analysis and (2) what you learned about this data set from your analysis. Write your response as if you were addressing a business manager (avoid using statistics jargon) and just provide the main take-aways.

In this data set we wanted to see if there was a correlation in life expectancy and PPGDP (countries wealth). We had different groups to help us determine if there was a trend between the different grouped countries. What we have learned is that there is a major connection between the wealth of a country having a impact on the life expectancy of a person in that country. That specific trend is the more wealthy a country usually means the higher life expectancy.

In [ ]: